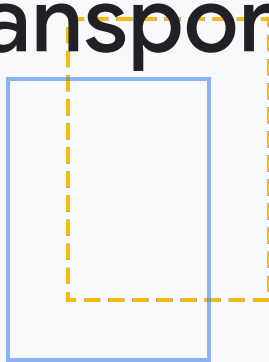


Stateless: the Future of MCP Transports

MCPcon Europe
Sept 17 2026

Shaun Smith, Hugging Face
Kurtis Van Gent, Google





Shaun Smith

**Hugging Face
Open Source/MCP & Agents**

**MCP Maintainer
Transport WG Lead**



Kurtis Van Gent

**Senior Staff Software Engineer
Google Cloud Databases**

**MCP Core Maintainer
Transport WG Lead**



MCP @ Scale

MCP on Google Cloud



Fully managed

No infrastructure to provision. Google handles hosting, scaling, and security.



Agent-first design

Built specifically for the reasoning patterns of modern LLMs.



Common governance for enterprise data

Managed service providing observability and governance required for enterprises.



Simplified discovery of data sources

Easily browse and access data sources through Gemini CLI and IDEs.



Data protection

Model Armor provides secure access management and identity control.

MCP Toolbox for Databases

Open Source & Extensible

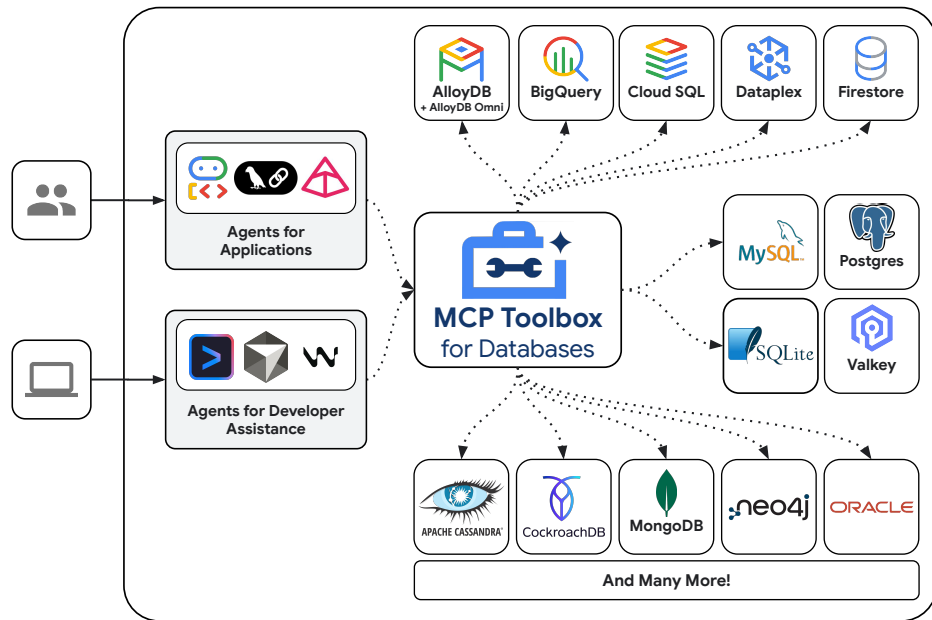
- ★ 16k+ Stars on GitHub, +175 contributors
- 🌐 40+ different databases

Highly Customizable

- Pre-built tools for common operations
- Custom tools for specific, narrow access for production-grade applications.**

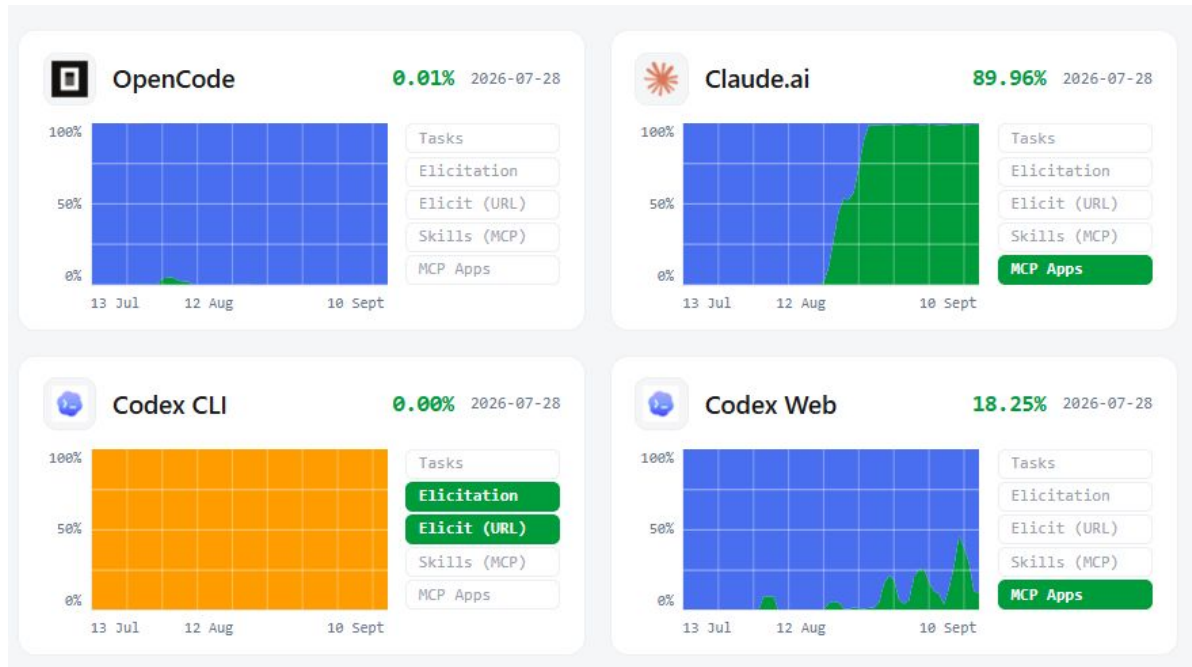
Scalable and Secure

- Secure connection management & credential handling
- Built-in connection pooling and OpenTelemetry support



MCP Adoption Dashboard

<https://huggingface.co/datasets/evalstate/mcp-clients>

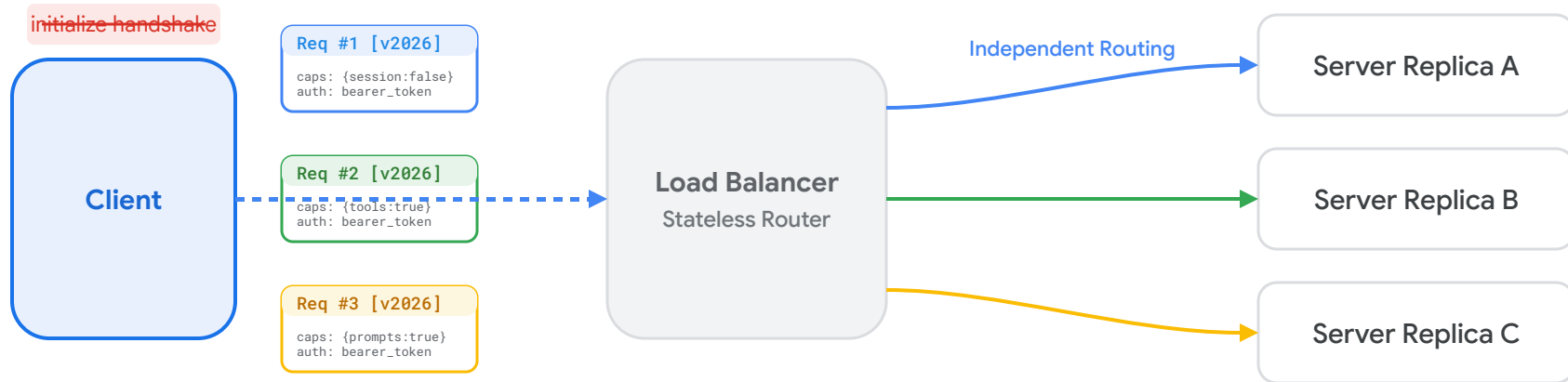


Open Dataset of Clients and Capabilities!

01

2026-07-28 highlights

SEP 2575: Stateless MCP



"Each request carries the protocol context needed to understand it."

Scale horizontally

Any server instance can handle any incoming request seamlessly.

Reduce coupling

Server failures don't drop user sessions or break active workflows.

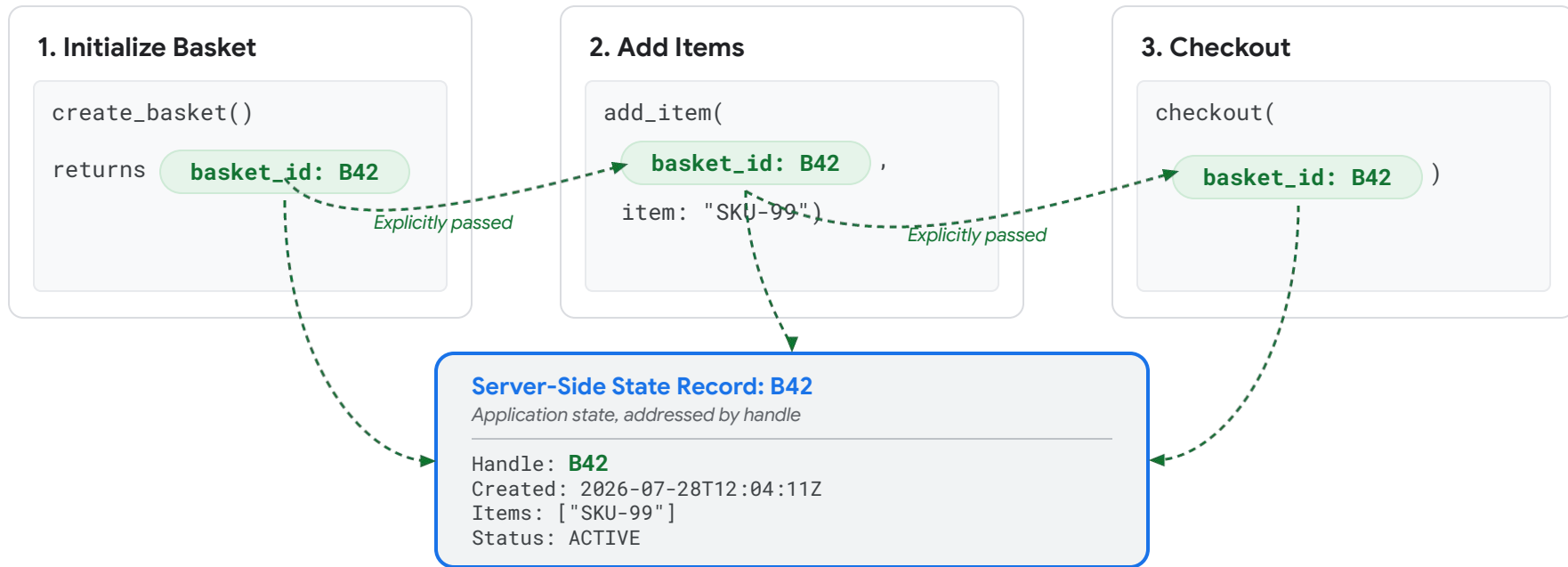
Simplify logic

Eliminate complex state synchronization across backend nodes.

SEP-2567: Sessionless MCP via Explicit State Handles

Each tool call must make its context explicit.

~~Map Session Id removed~~



The server does not infer context from earlier calls. Pass the inputs and state references each call needs.

MRTR - Stateless Elicitations

ROUND TRIP 1: INITIAL CALL - NO SERVER STATE

1. Tool Call

Client initiates request to Server to execute tool.

Client → Server



2. input_required

Server requires user input; returns `requestState` token.

+ STATE Token



CALL ENDS · Connection Closed

No pending call on Server. Client retains state handle locally until user responds.

ROUND TRIP 2: SUBMIT ELICITATION RESULT AND CONTEXT

3. User Elicitation

3a. "Confirm action?"

3b. User grants "Approved"

Client ↔ User



4. Resubmission Call

Client sends NEW call with `inputResponses` + original state handle.

Explicit State Handle



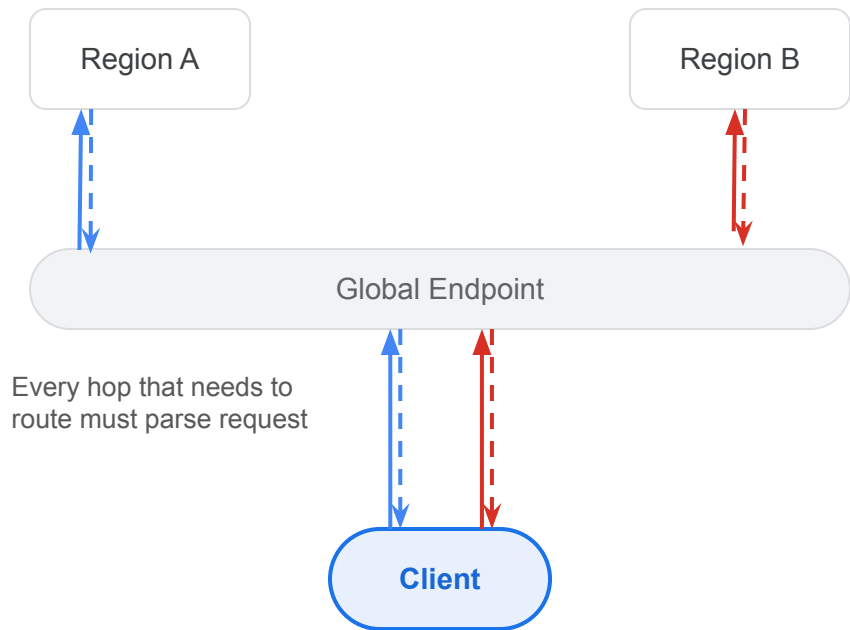
5. Complete Result

Server completes request and returns final result to Client.

Workflow Complete

SEP-2243: HTTP Standardization

S

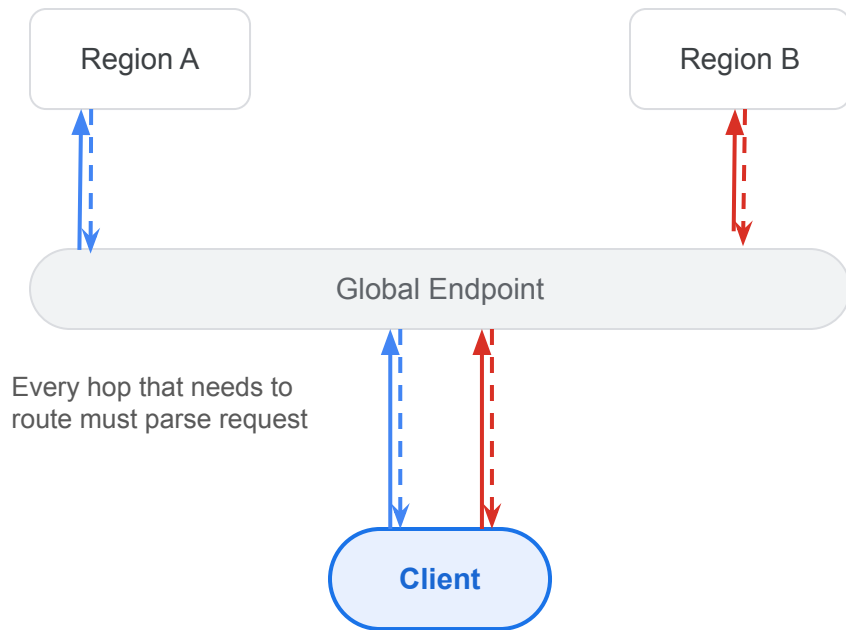


MCP Over HTTP

```
POST /mcp/ HTTP/1.1
Host: mcp-server.example
```

```
{
  "jsonrpc": "2.0",
  "method": "tools/call",
  "params": {
    "name": "spanner.execute_sql",
    "arguments": {
      "project": "senseai-prod",
      "region": "us-west1",
      "instance": "finance-db-01",
      "query": "SELECT *..."
    }
  }
}
```

SEP-2243: HTTP Standardization



MCP Over HTTP

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POST /mcp/ HTTP/1.1
Host: mcp-server.example
MCP-Name: spanner.execute_sql
MCP-Method: tools/call
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      "query": "SELECT *..."
    }
  }
}
```

SEP-2243: HTTP Standardization

Tool Definition (Schema)

```
{
  "name": "hf.run_in_sandbox",
  "inputSchema": {
    "properties": {
      "command": {
        "type": "string"
      },
      "sandbox_id": {
        "type": "string",
        "x-mcp-header": "Sandbox-Id"
      }
    }
  }
}
```

HTTP Request Headers **POST /mcp/**

```
Host: mcp-server.example
MCP-Name: hf.run_in_sandbox
MCP-Method: tools/call
X-MCP-Param-Sandbox-Id: sbx-292
X-MCP-Param-Region: us-west1
X-MCP-Param-Instance: finance-db-01
```

JSON-RPC Payload (Body)

```
{
  "jsonrpc": "2.0",
  "method": "tools/call",
  "params": {
    "name": "hf.run_in_sandbox",
    "arguments": {
      "Sandbox-Id": "sbx-292",
      "region": "us-west1",
      "instance": "finance-db-01",
      "cmd": "ls -al"
    }
  }
}
```

02

2027-3-31 ?

2026-12-15 Roadmap

Tool/Primitive Versioning

Problem

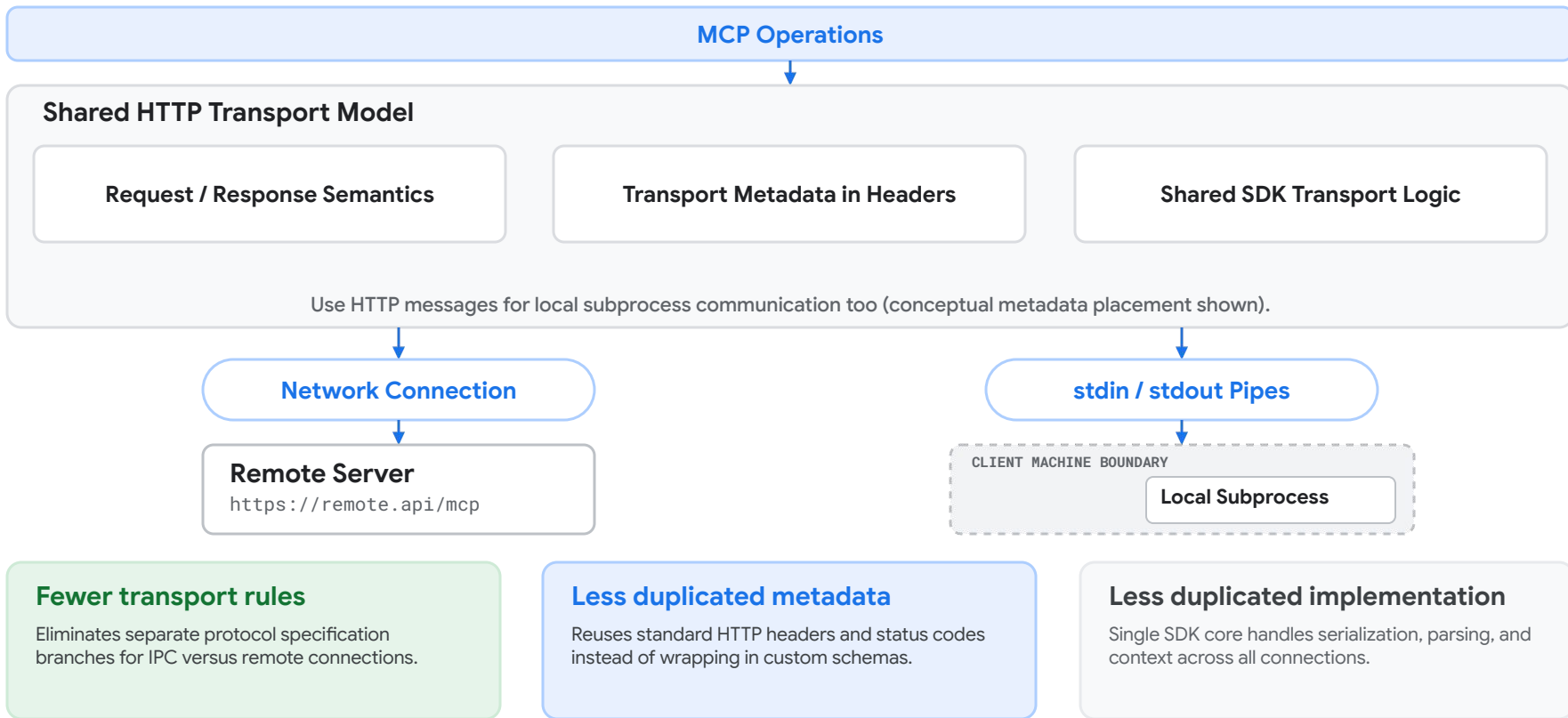
- **Stale Tool Cache:** A deployment can change a tool while clients still cache its old definition.
- **HTTP Cache Incompatibility:** JSON-RPC POST lists don't fit ordinary HTTP cache revalidation.

Solution

- **Per-tool digests:** Execute against the expected definition—or reject before execution and require a refresh.
- **Optional GET lists + ETags:** Reuse cached lists; return 304 Not Modified when unchanged.

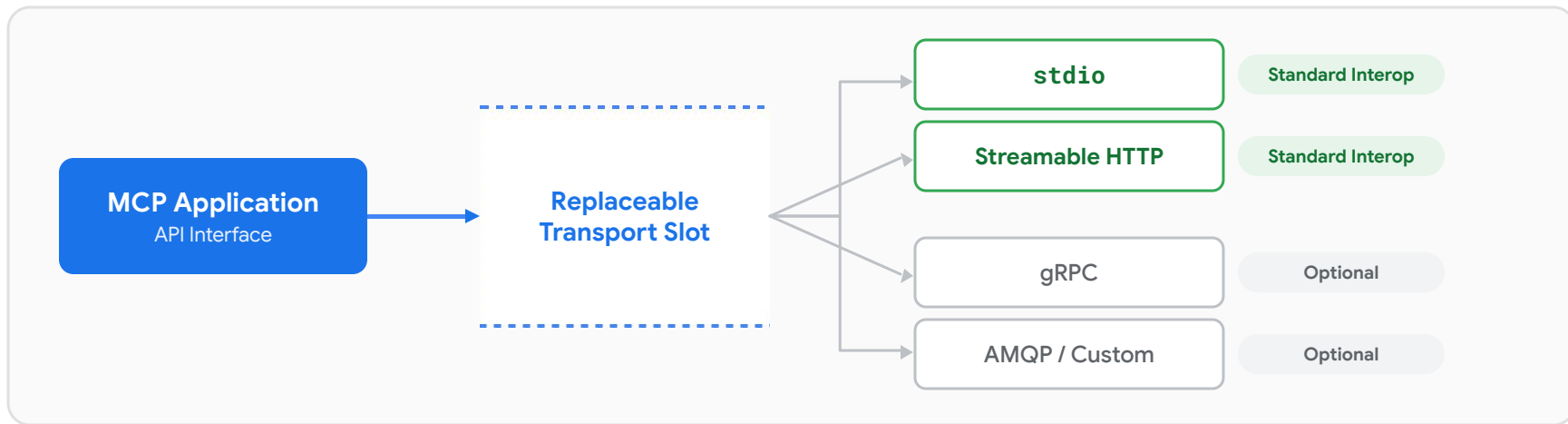
Opt-in proposals. Existing POST methods remain. Private lists stay private.

HTTP over STDIO



Pluggable Transports

“Change the transport, not the application.”



Decoupled Architecture

Application logic remains unchanged across local IPC and remote network boundaries.

Zero Runtime Leakage

Swap out transport implementations without updating client handler code.

Extensible Bindings

Add custom RPC or IPC transports by satisfying the common transport interface.

Best Effort

! Error Standardization

EXISTING ERROR EXAMPLE

MissingRequiredClientCapability

JSON-RPC: -32021

HTTP: 400 Bad Request

Missing capability: elicitation

PROPOSED ERROR STANDARDIZATION

Proposed

Error names

JSON-RPC codes

HTTP mappings

Retry guidance

Give clients a predictable way to respond. Server-side call-time validation exists today.

👁 Capability awareness

CACHED CATALOG CARD

confirm_booking

Requires elicitation

Proposed

Catalog entry remains valid and cached. Availability is dynamically re-evaluated based on active client capability state.

CLIENT CAPABILITY STATE CHANGE

Elicitation available

.....▶

Elicitation unavailable

MODEL TOOL EXCLUSION

~~**confirm_booking**~~

Excluded from model tools

Detect incompatibility before the model plans. Keep original catalog intact for later reuse.

Best Effort



Streaming Tool Results

Deliver partial results for UX/Observability, Steering and Efficiency



Content Language Negotiation

[Internationalization \(i18n\)](#)

Allow the Client to supply language preferences to update Human Facing Titles.

Capability

Dynamic Locale & Title Translation

Help shape the future of MCP transports

Join the MCP Transports Working Group.



Scan to get involved



[github.com/
modelcontextprotocol/transport-wg](https://github.com/modelcontextprotocol/transport-wg)

